

Early Identification of Heart Attack vs Gastritis Using AI/ML

R26-IT-012

Project Proposal Report

Jayalath J.R.D.S – IT22222640

**B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology**

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

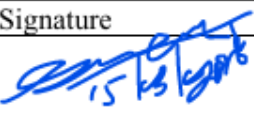
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Jayalath J.R.D.S	IT22222640	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
SMB Harshanath	15/03/2026	

ABSTRACT

Chest pain is one of the most frequently reported symptoms in emergency departments and primary healthcare settings, and differentiating between cardiac-related chest pain and non-cardiac causes such as gastric disorders remains a significant clinical challenge, particularly during early diagnosis. Gastric conditions including gastritis, gastroesophageal reflux disease (GERD), and peptic ulcer disease often produce symptoms that closely resemble heart attack symptoms, which can lead to misdiagnosis, delayed treatment, and unnecessary medical procedures. Although recent studies have applied artificial intelligence (AI) and machine learning techniques for cardiovascular disease detection and medical image analysis, most existing systems focus on either cardiac or gastrointestinal diseases independently and rarely integrate multiple diagnostic data sources such as patient symptoms, cardiac signals, and medical imaging within a single decision-support framework. To address this limitation, this research proposes an AI-assisted diagnostic system designed to differentiate cardiac and gastric causes of chest pain by integrating multiple clinical inputs. The proposed system consists of four main components: patient questionnaire analysis, cardiac signal analysis, coronary blockage detection, and gastric disease analysis. Component 4, the Gastric Analysis and Decision Support Engine, utilizes symptom triggers and gastroscopy images to identify gastric diseases using deep learning techniques. The system employs the EfficientNet-B4 convolutional neural network architecture to classify gastric diseases from endoscopy images, while image preprocessing is performed using OpenCV and Albumentations. Additionally, Grad-CAM is applied to provide visual explanations for model predictions, improving interpretability. A rule-based recommendation module then generates treatment suggestions based on the detected disease and severity level. The expected outcome of this research is an AI-driven clinical decision support system that enhances diagnostic accuracy and supports medical students in understanding the clinical reasoning process involved in chest pain diagnosis. From a societal perspective, the system has the potential to reduce misdiagnosis and improve patient management, while commercially it may support the development of healthcare decision-support platforms and medical education technologies.

Keywords: Artificial Intelligence, Gastric Disease Detection, Chest Pain Diagnosis, Medical Image Analysis, Clinical Decision Support.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our supervisor, Mr. Buddika Harshanath, for his continuous guidance, valuable feedback, and encouragement throughout the development of this research proposal. His insightful suggestions and academic support greatly contributed to shaping the direction and quality of this research work.

We are also deeply grateful to our co-supervisor, Prof. Samantha Rajapaksha, for his valuable advice, constructive comments, and continuous support during the planning and preparation of this research project.

We would like to extend our sincere appreciation to our external supervisors, Dr. Nishan Vas Gunawardhana and Dr. Hasintha from the Teaching Hospital – Polonnaruwa, for providing valuable clinical insights and professional guidance related to the medical aspects of this research. Their expertise and suggestions were extremely helpful in understanding the clinical perspective of chest pain diagnosis.

Our sincere thanks also go to the Sri Lanka Institute of Information Technology (SLIIT) and the Department of Information Technology for providing the academic environment, facilities, and resources required to carry out this research.

Finally, we would like to express our heartfelt gratitude to our parents, friends, and colleagues for their continuous support, encouragement, and motivation throughout our academic journey and during the completion of this research proposal.

TABLE OF CONTENT

DECLARATION	2
ABSTRACT.....	3
ACKNOWLEDGEMENT.....	4
TABLE OF CONTENT	5
LIST OF TABLES	7
LIST OF FIGURES	8
LIST OF ABBREVIATIONS.....	9
LIST OF APPENDICES	10
1. INTRODUCTION	11
1.1. Background and Context	11
1.2. Literature Review	12
1.3. Research Gap.....	13
2. OBJECTIVE	16
2.1. Main Objective	16
2.2. Specific Objectives	16
3. METHODOLOGY.....	17
3.1. Research Approach	17
3.2. Technical Methods and Algorithms.....	17
3.3. Tools and Platforms	18
3.4. Data Collection and Ethical Considerations.....	18
3.5. Validation Strategy and Performance Metrics	19
4. HIGH-LEVEL SYSTEM ARCHITECTURE.....	19
4.1. Overview of Proposed Solution and Its Components	19
4.2. Overall System Diagram	20
4.3. High-Level Architecture of Gastric Analysis and Decision Support Engine .	20
4.4. Component Interactions	21
4.5. Data Flow Explanation	22
4.6. Scalability Considerations.....	23
4.7. Security Considerations	24
4. USER REQUIREMENTS.....	24
4.1. Stakeholder Identification	24

5.2.	Functional Requirements	25
5.3.	Non-Functional Requirements.....	25
6.	COMMERCIALIZATION PLAN	26
6.1.	Target Market and Customer Persona	26
6.2.	Value Proposition.....	27
6.3.	Revenue Model.....	27
6.4.	Cost Estimation	27
6.5.	Pricing Strategy.....	27
6.6.	Competitive Advantage	28
6.7.	Intellectual Property Considerations	28
7.	BUDGET AND JUSTIFICATION	28
7.1.	Estimated Budget Breakdown	28
7.2.	Hardware Cost Justification.....	29
7.3.	Software Requirements.....	29
7.4.	Cloud Services	30
7.5.	Human Effort Cost	30
7.	WORK BREAKDOWN STRUCTURE (WBS).....	31
8.1.	Work Breakdown Structure Diagram.....	31
8.2.	Task Breakdown per Member	32
8.3.	Timeline Alignment	32
8.4.	Balanced Workload Distribution	33
9.	GANTT CHART	34
	References.....	35
10.	APPENDICES.....	36

LIST OF TABLES

Table 1 : List of Abbreviations.....	9
Table 2 : List of Appendices	10
Table 3 : Comparison of Existing Research Studies and the Proposed Diagnostic System ...	15
Table 4 : Functional Requirements	25
Table 5 : Budget Breakdown for the Gastric Analysis and Decision Support Engine	29
Table 6 : Software Requirements for System Development	30
Table 7 : Cloud Infrastructure Cost.....	30
Table 8 : Human Effort Cost.....	31
Table 9 : Task Breakdown per Member	32
Table 10 : Timeline Alignment.....	33
Table 11 : Risk Analysis	36

LIST OF FIGURES

Figure 1 : High-Level System Architecture Diagram.....	20
Figure 2 : Gastric Analysis and Decision Support Engine - Component Architecture.....	21
Figure 3 : Gastric Analysis and Decision Support Engine - Component Interaction Diagram	22
Figure 4 : Gastric Analysis and Decision Support Engine - Data Flow Diagram	23
Figure 5 : WORK BREAKDOWN STRUCTURE	31
Figure 6 : Gantt Chart	34

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
ECG	Electrocardiogram
GERD	Gastroesophageal Reflux Disease
CDSS	Clinical Decision Support System
DSR	Design Science Research
API	Application Programming Interface
REST	Representational State Transfer
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
F1-score	Harmonic Mean of Precision and Recall
IDE	Integrated Development Environment
GPU	Graphics Processing Unit
SQL	Structured Query Language
UI	User Interface
Grad-CAM	Gradient-weighted Class Activation Mapping
SaaS	Software as a Service

Table 1 : List of Abbreviations

LIST OF APPENDICES

Appendix	Title
Appendix A	Survey Instruments
Appendix B	Preliminary UI Mockups
Appendix C	Risk Analysis
Appendix D	Ethical Documents

Table 2 : List of Appendices

1. INTRODUCTION

1.1. Background and Context

Chest pain is one of the most frequently reported symptoms in emergency departments and primary healthcare settings worldwide. It can arise from various medical conditions, including life-threatening cardiac diseases such as myocardial infarction and angina, as well as non-cardiac causes, including gastrointestinal disorders. In clinical practice, distinguishing between cardiac and non-cardiac chest pain remains a challenging task because several gastric conditions, such as gastritis, gastroesophageal reflux disease (GERD), and peptic ulcer disease, may produce symptoms that closely resemble cardiac chest pain [1]. This overlap in symptoms often leads to diagnostic uncertainty, especially during early clinical evaluation.

Accurate and timely identification of the underlying cause of chest pain is critical because delays in diagnosing cardiac conditions can result in serious complications or even mortality. At the same time, misclassification of gastric-related chest pain as cardiac events can lead to unnecessary medical investigations, increased healthcare costs, and patient anxiety. Traditional diagnostic procedures typically involve patient history analysis, electrocardiogram (ECG) testing, and cardiac biomarker evaluations such as troponin levels to detect potential cardiac abnormalities [2]. Although these diagnostic approaches are effective in identifying cardiac issues, they may not always provide sufficient information to differentiate cardiac conditions from gastrointestinal disorders during the initial stages of diagnosis.

Recent advancements in artificial intelligence (AI) and machine learning have introduced new opportunities for improving clinical decision-making. AI-based systems can analyze large volumes of patient data, physiological signals, and medical imaging information to assist healthcare professionals in identifying diseases more accurately and efficiently. Machine learning algorithms such as Random Forest, Support Vector Machines, and Gradient Boosting models have been successfully applied in cardiovascular disease prediction using patient demographic data and clinical indicators [3]. Similarly, deep learning models, particularly convolutional neural networks (CNNs), have demonstrated high performance in analyzing medical signals such as ECG recordings and detecting cardiac abnormalities with high accuracy [4].

In addition to cardiac diagnostics, artificial intelligence has also been widely applied in gastrointestinal disease detection. Gastroscopy or endoscopy is considered the most reliable method for diagnosing gastric disorders, including gastritis, ulcers, and other gastrointestinal abnormalities. However, interpreting endoscopic images often requires experienced specialists and can be time-consuming. Recent studies indicate that deep learning models can automatically detect abnormalities in gastrointestinal images, improving diagnostic accuracy and reducing the workload of clinicians [5] [6].

Despite these technological advancements, most existing diagnostic systems focus on analyzing either cardiac or gastrointestinal conditions separately. Few systems integrate multiple clinical inputs, such as patient symptoms, ECG results, and medical imaging data, into a single decision-support framework. Therefore, there is a growing need for an intelligent

clinical decision support system that can assist medical students and healthcare professionals in differentiating cardiac-related chest pain from gastric-related chest pain. The proposed research addresses this need by developing an AI-assisted diagnostic system that integrates patient questionnaires, cardiac signal analysis, and gastric disease detection. Within this framework, Gastric Analysis and Decision Support Engine focuses on analyzing gastric symptom triggers and endoscopy images using deep learning techniques to identify gastric conditions that may mimic cardiac chest pain and support accurate clinical decision-making.

1.2.Literature Review

The increasing use of artificial intelligence (AI) and machine learning in healthcare has significantly improved disease detection, medical imaging analysis, and clinical decision-making. In recent years, several studies have explored the use of machine learning techniques to assist healthcare professionals in diagnosing cardiovascular diseases and other medical conditions. Early detection of cardiac diseases is critical because cardiovascular diseases remain one of the leading causes of mortality worldwide. Researchers have therefore investigated various predictive models that analyze patient clinical data, physiological signals, and medical history to improve diagnostic accuracy.

Several studies have applied machine learning algorithms such as Random Forest, Support Vector Machines, and Gradient Boosting models for cardiovascular disease prediction. These models analyze risk factors, including age, blood pressure, cholesterol levels, and lifestyle habits to predict the likelihood of heart-related diseases [3]. Research has shown that machine learning-based predictive models can achieve higher accuracy compared to traditional statistical approaches, providing healthcare professionals with valuable decision support during diagnosis [7] [8].

Deep learning techniques have also been widely used in medical signal analysis, particularly in electrocardiogram (ECG) interpretation. Convolutional Neural Networks (CNNs) and recurrent neural networks have demonstrated strong performance in detecting cardiac abnormalities such as arrhythmias and myocardial infarction from ECG signals. For example, research conducted by Hannun et al. demonstrated that deep neural networks can achieve cardiologist-level performance in detecting arrhythmias using large-scale ECG datasets [4]. These advancements highlight the potential of AI-based systems in supporting early cardiac diagnosis and reducing diagnostic errors.

In addition to cardiac disease detection, artificial intelligence has been increasingly applied in the analysis of gastrointestinal diseases. Gastroscopy and endoscopy are considered the most reliable diagnostic methods for identifying gastric disorders such as gastritis, ulcers, and gastrointestinal inflammation. However, manual analysis of endoscopic images requires extensive clinical expertise and may be time-consuming. Recent studies have shown that deep learning models can automatically detect abnormalities in gastrointestinal images, improving diagnostic efficiency and accuracy [5] [6].

Modern deep learning architectures such as **EfficientNet** have demonstrated high performance in image classification tasks while maintaining computational efficiency. EfficientNet models

use compound scaling techniques that optimize network depth, width, and resolution simultaneously, enabling the model to achieve higher accuracy with fewer parameters compared to traditional convolutional neural networks [9]. Because of these advantages, EfficientNet architectures have been successfully applied in various medical imaging applications, including endoscopic image analysis [6] [10].

Furthermore, several studies have investigated the development of clinical decision support systems (CDSS) that integrate artificial intelligence into healthcare workflows. These systems analyze patient information, medical test results, and diagnostic images to provide recommendations that assist clinicians in making informed decisions. AI-based decision support systems have been widely adopted in medical fields such as radiology, cardiology, and pathology to improve diagnostic accuracy and reduce clinical workload [11] [12].

Despite these technological advancements, most existing research focuses on either cardiac disease detection or gastrointestinal disease analysis independently. Few systems integrate multiple diagnostic sources such as patient symptoms, physiological signals, and medical imaging into a unified diagnostic framework. Additionally, many existing AI-based diagnostic tools are primarily designed to assist experienced clinicians rather than to support the clinical learning process of medical students.

Therefore, there is a growing need for an integrated AI-assisted clinical decision support system that can analyze multiple clinical inputs to differentiate between cardiac-related chest pain and gastric-related chest pain. The proposed research aims to address this gap by integrating patient questionnaires, ECG analysis, and gastric disease detection into a unified diagnostic workflow. In particular, Component 4 focuses on gastric analysis using symptom triggers and endoscopy image classification with EfficientNet-based deep learning models to identify gastric conditions that may mimic cardiac chest pain and support accurate medical decision-making.

1.3. Research Gap

A detailed review of existing research and clinical diagnostic systems reveals several limitations in current approaches used for chest pain diagnosis and medical decision support. Although many studies have explored artificial intelligence in cardiovascular disease detection and gastrointestinal disease analysis, there is still a lack of integrated systems capable of differentiating cardiac-related chest pain from gastric-related chest pain. These limitations highlight the need for an intelligent AI-assisted clinical decision support system that can support medical students and healthcare professionals in making accurate diagnostic decisions.

➤ **Limited Integration Between Cardiac and Gastric Diagnosis**

Most existing medical AI research focuses on either cardiovascular disease detection or gastrointestinal disease analysis independently. Many studies have developed machine learning models for detecting cardiac abnormalities from ECG signals or predicting cardiovascular risk factors. Similarly, other studies focus on identifying gastric diseases using endoscopy images. However, these systems are designed for single-disease analysis and do not address the clinical challenge of differentiating between cardiac and gastric causes of chest pain. As a result, there

is a lack of integrated diagnostic frameworks that analyze both cardiac and gastric indicators simultaneously.

➤ **Over-Reliance on Single Data Sources**

Many existing diagnostic systems rely on a single type of clinical data, such as ECG signals, patient demographic information, or medical images. While these approaches provide valuable insights, real clinical diagnosis requires the integration of multiple data sources including patient symptoms, laboratory results, physiological signals, and medical imaging. Systems that rely on a single data modality may fail to capture the complete clinical picture, which can reduce diagnostic accuracy. Therefore, there is a need for a multi-modal AI system that integrates patient questionnaires, cardiac testing, and gastric imaging analysis.

➤ **Limited Application of AI in Gastric Chest Pain Analysis**

Although AI has been widely used in cardiac disease detection, relatively fewer studies focus on gastric conditions that can mimic cardiac chest pain. Gastric disorders such as gastritis, GERD, and peptic ulcers can produce symptoms similar to heart-related chest pain, which often leads to misdiagnosis in clinical settings. Existing research primarily focuses on detecting gastric abnormalities in endoscopy images but rarely connects these findings with symptom-based diagnostic reasoning or clinical decision support systems.

➤ **Lack of Educational Decision Support Tools for Medical Students**

Most existing AI-based diagnostic systems are developed to assist experienced clinicians rather than to support the learning process of medical students. These systems typically focus on automated disease detection without explaining the reasoning behind diagnostic decisions. Medical students require systems that not only identify potential diseases but also demonstrate the clinical reasoning process behind differential diagnosis. However, there are limited AI systems designed specifically to support medical education and diagnostic training.

➤ **Absence of Integrated Treatment Recommendation Systems**

Many current AI diagnostic systems focus mainly on disease detection or classification tasks. While disease identification is important, it does not provide complete clinical support for decision-making. Healthcare professionals and medical students require systems that can also generate treatment recommendations based on the detected condition and its severity. Existing systems rarely integrate diagnostic prediction with automated treatment planning or prescription generation modules.

➤ **Lack of Comprehensive AI-Based Clinical Decision Frameworks**

Although several AI models exist for medical diagnosis, few studies combine multiple diagnostic components into a unified clinical workflow. Real-world medical diagnosis often involves sequential steps, including symptom evaluation, diagnostic testing, imaging analysis, and treatment planning. However, most current research focuses on isolated AI models rather than comprehensive decision support frameworks that simulate real clinical workflows.

Study	Heart Prediction	Cardiac vs Gastric	Early Screening	AI Decision	Education Support	Integrated Workflow
Pervez et al. [1]	✓	✓	✗	✗	✗	✗
Predictive Analytics [2]	✓	✗	✗	✓	✗	✗
Mobile Cardiac Monitor [3]	✓	✗	✓	✗	✗	✗
Atypical MI Study [4]	✓	✓	✗	✗	✗	✗
Takotsubo Study [5]	✓	✓	✗	✗	✗	✗
Blockchain Health System [6]	✗	✗	✗	✗	✗	✗
Vascular Vision Learning [7]	✗	✗	✗	✗	✓	✗
ML Heart Prediction [8]	✓	✗	✗	✓	✗	✗
DL ECG Detection [9]	✓	✗	✗	✓	✗	✗
Clinical Decision System [10]	✓	✗	✗	✓	✗	✗
Proposed System	✓	✓	✓	✓	✓	✓

Table 3 : Comparison of Existing Research Studies and the Proposed Diagnostic System

In this table:

- ✓ – Feature supported
- ✗ – Feature not supported

2. OBJECTIVE

2.1.Main Objective

The main objective of this research is to develop a Gastric analysis and Decision Support module that analyzes patient symptoms and endoscopy images to accurately identify gastric-related causes of chest pain and generate appropriate treatment recommendations, thereby supporting medical students in differentiating gastric chest pain from cardiac conditions.

2.2.Specific Objectives

➤ **Symptom Pattern Analysis for Gastric Disease Identification**

The first objective of this research is to analyze gastric-related symptoms collected through the patient questionnaire in order to identify symptom patterns associated with gastric diseases. Symptoms such as upper abdominal pain, burning sensation after meals, and relief after antacid medication can indicate gastrointestinal disorders. Analyzing these symptoms helps identify gastric conditions that may mimic cardiac chest pain.

➤ **Development of an AI-Based Gastric Disease Detection Model**

Another objective is to develop an artificial intelligence model capable of detecting gastric diseases using endoscopy images. Deep learning techniques, particularly convolutional neural networks, can identify abnormalities such as gastric inflammation and ulcers. In this research, the EfficientNet-B4 architecture is used to classify gastric diseases from gastroscopy images.

➤ **Integration of Symptom-Based and Image-Based Diagnosis**

This research also aims to integrate symptom-based predictions with AI-based image analysis to improve diagnostic accuracy. Symptom analysis alone may not provide reliable diagnosis because several gastric diseases share similar symptoms. Therefore, combining symptom data with endoscopy image analysis helps confirm the final gastric diagnosis.

➤ **Automatic Generation of Treatment Recommendations**

Another objective of the system is to generate treatment recommendations based on the detected gastric disease and its severity level. The system provides medication suggestions, dosage recommendations, and dietary advice to support patient management.

➤ **Educational Decision Support for Medical Students**

The final objective of this research is to support medical students in understanding the diagnostic process of gastric diseases. The system provides explanations and recommendations that help students understand the relationship between symptoms, diagnostic tests, and treatment decisions.

3. METHODOLOGY

The proposed system follows an AI-driven clinical decision support approach to assist medical students in differentiating between cardiac-related and gastric-related chest pain. The overall methodology integrates patient symptom analysis, cardiac diagnostic testing, and gastric disease detection within a structured diagnostic workflow. This section describes the research approach, technical methods, tools and technologies, data collection process, and validation strategies used in the development of the system.

Within the overall system, Gastric Analysis & Decision Support Engine is responsible for analyzing gastric symptoms and endoscopy images to confirm gastric diseases and generate treatment recommendations.

3.1. Research Approach

This research follows a Design Science Research (DSR) approach, which focuses on designing and developing an intelligent artifact to solve a real-world problem. The artifact developed in this research is an AI-assisted clinical decision support system that integrates multiple diagnostic components to simulate real clinical workflows.

The proposed system processes patient questionnaire data, cardiac diagnostic test results, and gastric imaging data through several AI models. If cardiac causes are excluded through earlier diagnostic components, the system activates Component 4, which performs gastric analysis using symptom triggers and endoscopy image classification. The component then generates diagnostic results, severity levels, and treatment recommendations to assist medical students in making informed clinical decisions.

3.2. Technical Methods and Algorithms

The proposed system utilizes a combination of machine learning and deep learning techniques to analyze different types of medical data.

For the gastric diagnostic module, deep learning models are used to analyze endoscopy images to identify abnormalities such as gastritis, gastric ulcers, and mucosal inflammation. The system applies the EfficientNet-B4 convolutional neural network architecture, which has demonstrated high performance in medical image classification tasks. Transfer learning is used to improve model performance by leveraging pretrained weights and adapting them to the gastric disease dataset.

Medical images are preprocessed using image processing techniques to improve model performance. Image normalization, resizing, and augmentation are applied to increase dataset diversity and improve generalization. Data augmentation techniques such as rotation, flipping, and contrast adjustment are applied using the Albumentations library.

To improve interpretability, Grad-CAM (Gradient-weighted Class Activation Mapping) is used to highlight important regions in endoscopy images that influence the model's predictions. This allows medical students to visually understand how the AI model identifies gastric abnormalities.

In addition to image-based diagnosis, a rule-based recommendation layer is implemented to generate treatment suggestions based on the detected disease and severity level. This layer maps diagnostic outcomes to appropriate medication recommendations, dietary guidance, and clinical management suggestions.

3.3.Tools and Platforms

The system is implemented using a modern AI development stack designed for medical data analysis and web-based decision support systems.

The backend of the system is developed using Python, which is widely used for machine learning and medical imaging applications. FastAPI is used as the backend framework to connect AI models with the user interface through RESTful APIs. FastAPI enables efficient communication between the frontend application and the AI models.

The deep learning models are developed using the PyTorch framework, which provides flexibility for implementing convolutional neural networks and transfer learning techniques. OpenCV is used for medical image preprocessing and image manipulation, while Albumentations is used for performing image augmentation during model training.

Patient data, questionnaire responses, prediction results, and system logs are stored in a PostgreSQL database, which provides reliable storage for structured medical data.

The frontend interface for medical students is developed using React.js, which allows the creation of interactive dashboards for displaying diagnostic results, AI predictions, and visual explanations. Tailwind CSS or Material UI is used to design the user interface.

To support AI model experimentation and performance monitoring, MLflow is used for tracking model training experiments and evaluation results. The system is deployed using Docker containers to ensure portability and consistent execution across different computing environments.

3.4.Data Collection and Ethical Considerations

The dataset used for training the gastric disease detection model consists of gastroscopy or endoscopy images representing different gastric conditions such as gastritis, ulcers, and normal gastric mucosa. These images may be obtained from publicly available medical imaging datasets or approved clinical data sources.

Before training the AI models, all datasets are preprocessed to ensure data quality and consistency. Personal identifiers and sensitive patient information are removed to ensure patient privacy and confidentiality.

The research follows ethical guidelines for medical data usage, ensuring that all patient data is anonymized and used solely for research and educational purposes.

3.5. Validation Strategy and Performance Metrics

To evaluate the performance of the gastric disease detection model, the dataset is divided into training, validation, and testing sets. The trained model is evaluated using several performance metrics commonly used in medical image classification.

Key evaluation metrics include accuracy, precision, recall, and F1-score, which measure the model's ability to correctly identify gastric diseases. In addition, confusion matrices are used to analyze classification performance across different disease categories.

For image classification tasks, Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) may also be used to evaluate the diagnostic performance of the model.

The explainability component using Grad-CAM is also evaluated qualitatively by verifying whether highlighted image regions correspond to clinically relevant gastric abnormalities.

The final system will be tested using simulated clinical scenarios to ensure that the AI predictions, severity classifications, and treatment recommendations align with expected medical reasoning. This validation process ensures that the proposed decision support system provides reliable and interpretable outputs suitable for educational and clinical support purposes.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1. Overview of Proposed Solution and Its Components

The proposed research system is an AI-assisted clinical decision support platform designed to differentiate cardiac-related chest pain from gastric-related chest pain through a structured diagnostic workflow. The complete solution consists of four major components that work together to simulate real clinical reasoning and support medical student learning.

Component 1 – Baseline Risk & Patient Profile Analysis performs the initial screening stage using questionnaire data, patient demographics, medical history, lifestyle factors, and early biomarkers. It calculates cardiac and gastric risk probabilities and categorizes the patient into the appropriate preliminary pathway.

Component 2 – Heart Signal & Functional Analysis processes ECG and echocardiography data to detect cardiac abnormalities. This component supports confirmation of suspected cardiac conditions through AI-assisted signal and image analysis.

Component 3 – Coronary Blockage Detection analyzes coronary angiography data to identify arterial blockage location, severity, and treatment direction. This component supports final cardiac pathway decision-making.

Component 4 – Gastric Analysis & Decision Support Engine is activated only when ECG and Troponin results indicate that the patient belongs to the gastric pathway. This module receives symptom triggers from Component 1 together with gastroscopy/endoscopy images, applies image preprocessing and EfficientNet-B4-based analysis, confirms the gastric disease,

determines severity, and generates treatment recommendations. The component also supports medical students by providing diagnostic explanations and feedback.

4.2.Overall System Diagram

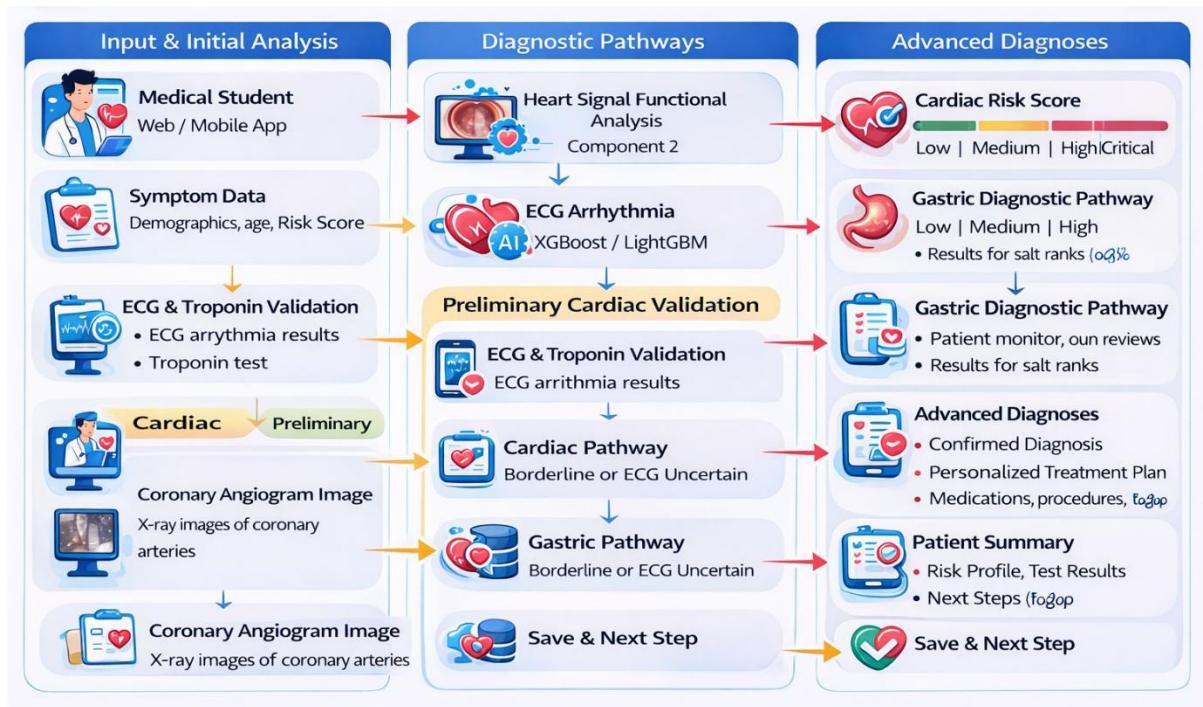


Figure 1 : High-Level System Architecture Diagram

4.3.High-Level Architecture of Gastric Analysis and Decision Support Engine

The Gastric Analysis & Decision Support Engine is responsible for diagnosing gastric-related causes of chest pain after cardiac conditions have been ruled out. The component integrates symptom-based reasoning and AI-based endoscopy image analysis to produce a final gastric diagnosis and treatment recommendation. According to the workflow, this component analyzes symptom triggers such as meal-related pain, spicy food aggravation, and relief after antacid use, together with gastroscopy images, and classifies gastric severity as Low, Medium, or Critical.

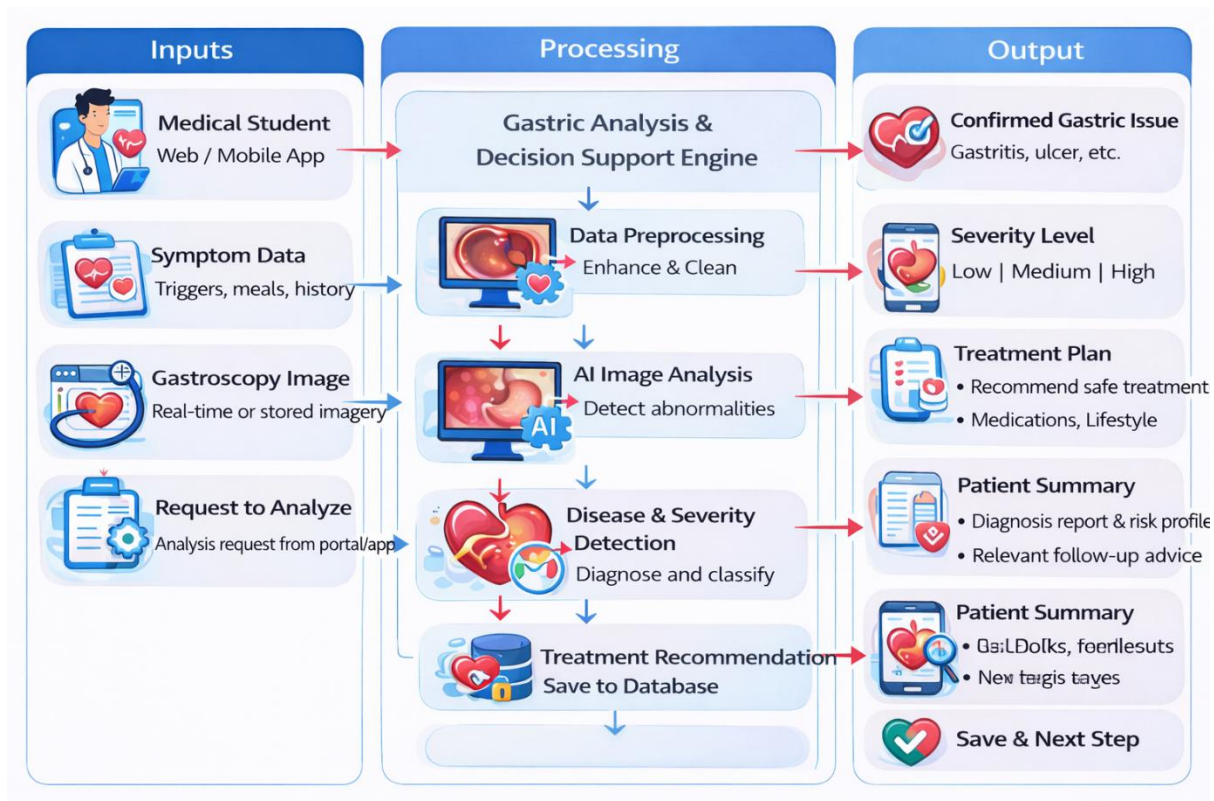


Figure 2 : Gastric Analysis and Decision Support Engine - Component Architecture

4.4.Component Interactions

Component 4 interacts with both earlier and later stages of the overall diagnostic system.

Interaction with Component 1:

Component 4 receives the patient symptom profile, gastric risk probability, and symptom triggers generated by the initial screening component. These inputs guide the first-stage gastric reasoning process.

Interaction with Diagnostic Imaging Sources:

The component receives gastroscopy or endoscopy images captured during the gastric examination stage. Optional upper abdominal ultrasound data may also be used when available to support diagnosis.

Interaction with Final Clinical Output Layer:

After analysis, the component provides the confirmed gastric diagnosis, severity level, AI-generated treatment plan, and student-oriented feedback to the final output layer of the system. This output supports both clinical interpretation and educational decision support.

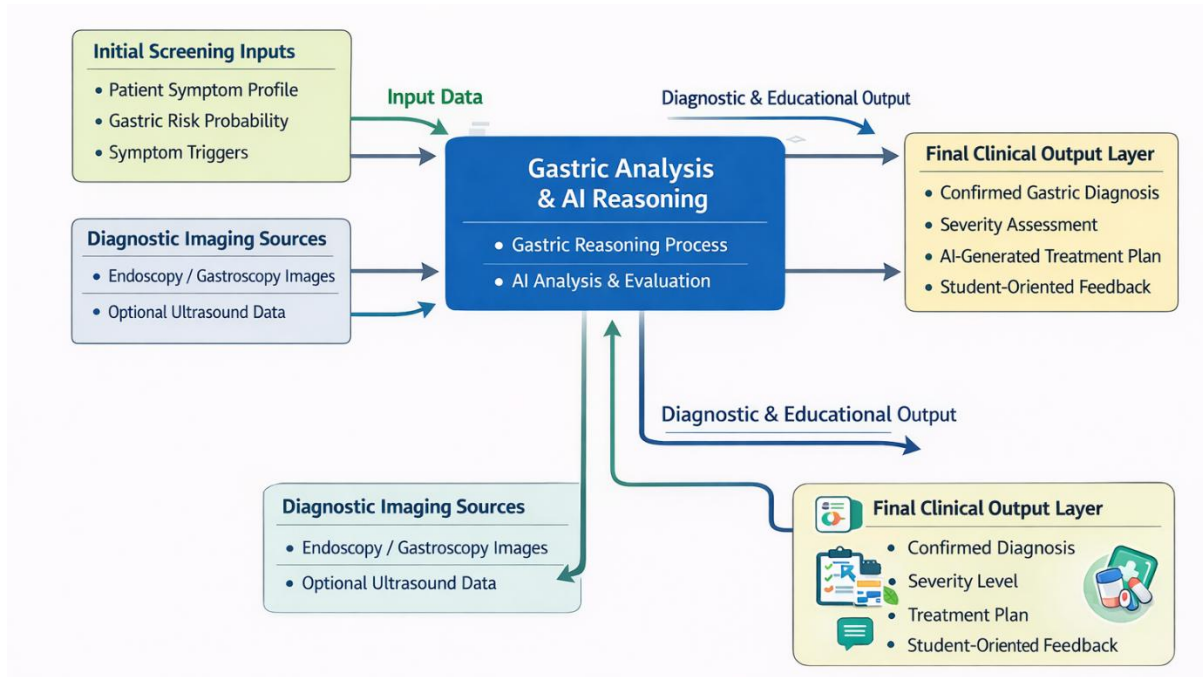


Figure 3 : Gastric Analysis and Decision Support Engine - Component Interaction Diagram

4.5.Data Flow Explanation

The data flow of Component 4 follows a structured sequence:

Step 1 – Input Reception:

The module receives symptom triggers and gastric prediction context from Component 1, including features such as pain related to meals, spicy food triggers, pain during fasting, and relief after antacid medication.

Step 2 – Initial Gastric Prediction:

Based on symptom patterns, the system performs an initial gastric disease prediction. Possible outcomes include gastritis, gastroesophageal reflux disease (GERD), peptic ulcer, and hiatal hernia.

Step 3 – Image Acquisition and Preprocessing:

Gastroscopy or endoscopy images are captured and prepared for AI analysis. Preprocessing includes image resizing, normalization, and noise removal to improve model performance.

Step 4 – AI Image Analysis:

The preprocessed images are passed to the EfficientNet-B4 deep learning model, which extracts image features and detects abnormalities such as inflammation, ulcers, mucosal irritation, and related gastric findings.

Step 5 – Disease Confirmation and Severity Detection:

The system combines symptom-based prediction with AI image classification to confirm the final gastric disease. It then determines the severity level as **Low, Medium, or Critical**.

Step 6 – Treatment Recommendation Generation:

A rule-based recommendation layer maps the detected disease and severity level to medication guidance, dietary advice, and treatment suggestions.

Step 7 – Final Output Delivery:

The module returns the confirmed diagnosis, severity classification, treatment recommendation, and student feedback, which can be displayed through the medical student dashboard.

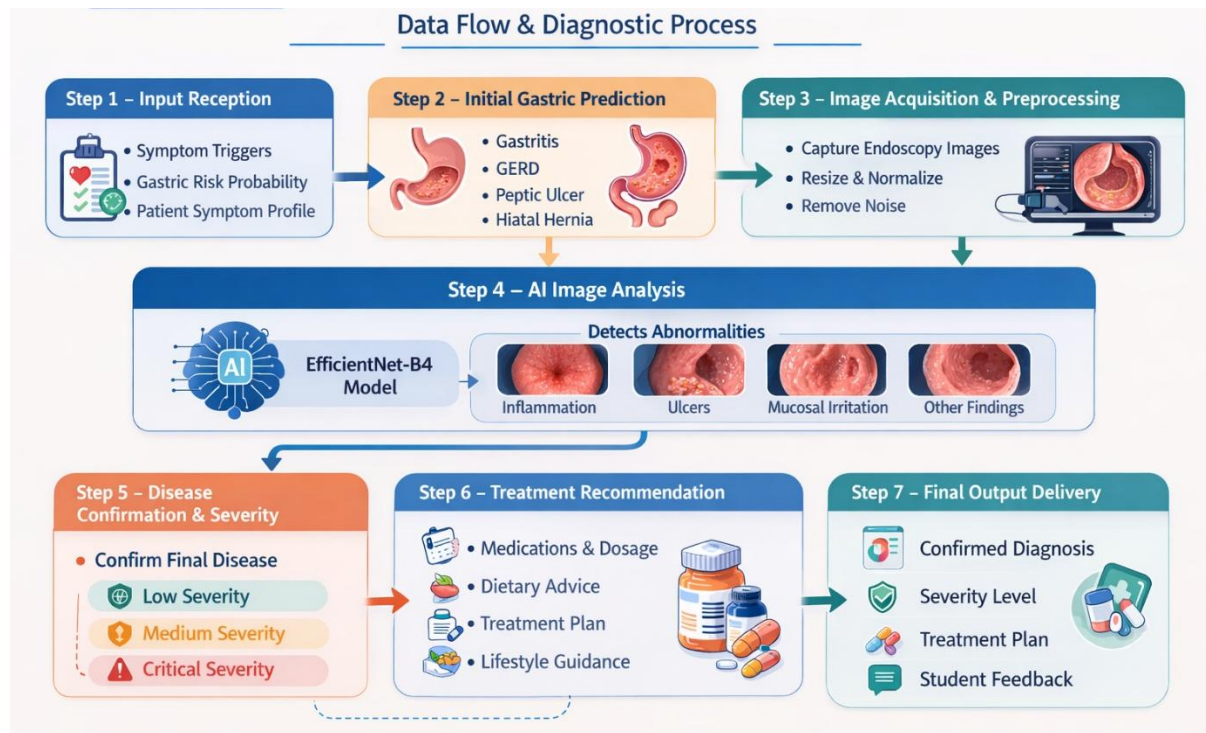


Figure 4 : Gastric Analysis and Decision Support Engine - Data Flow Diagram

4.6. Scalability Considerations

The architecture of Component 4 is designed to support future scalability.

First, the module follows a modular AI architecture, allowing the gastric analysis engine to operate independently while integrating with the other system components. This makes it easier to update or replace the AI model without redesigning the entire system.

Second, the backend can be implemented using Python and FastAPI, which provides lightweight API communication and allows the component to be connected with web-based dashboards or other services.

Third, the system can support future expansion by adding:

- More gastric disease classes,
- More advanced explainable AI visualizations,
- Additional imaging sources,
- Larger training datasets for improved generalization.

4.7. Security Considerations

Since the component handles medical images and diagnostic outputs, security is an important architectural concern.

The system should ensure that all uploaded endoscopy images and diagnostic results are stored securely and accessed only by authorized users. Patient information should be anonymized so that personally identifiable data is not stored within the analysis pipeline. This aligns with the privacy-preserving direction already described in the broader system design.

In addition, secure backend communication, controlled access to the dashboard, and protected database storage should be used to reduce the risk of unauthorized access or data leakage. PostgreSQL-based structured storage and secure API-based communication are appropriate for this type of architecture.

4. USER REQUIREMENTS

4.1. Stakeholder Identification

Stakeholders are individuals or groups who interact with the system, contribute to its development, or benefit from its outputs. The Gastric Analysis & Decision Support Engine involves several stakeholders, including medical students, patients, researchers, and system developers.

Medical Students

Medical students are the primary users of this component. The system assists them in understanding how gastric diseases can cause symptoms similar to cardiac chest pain. Students can review endoscopy images, observe AI-based disease predictions, and learn how gastric conditions such as gastritis, ulcers, or GERD are identified using medical imaging. The system also generates automated prescription suggestions, helping students learn appropriate treatment strategies.

Patients

Patients contribute indirectly by providing symptom information and undergoing diagnostic procedures such as gastroscopy. The system analyzes anonymized patient data including symptom triggers and endoscopy images, to detect potential gastric diseases. Patients benefit from the system through improved diagnostic support and more accurate identification of gastric causes of chest pain.

Researchers

Researchers analyze the performance of the gastric disease detection model and evaluate the effectiveness of AI-based image classification methods. They use anonymized endoscopy datasets and system outputs to assess model accuracy, study medical image analysis techniques, and improve the diagnostic algorithms.

Developers

Developers are responsible for implementing the gastric analysis module. Their responsibilities include developing deep learning models for endoscopy image classification, implementing backend APIs, integrating the component with other system modules, and maintaining system performance and security.

5.2.Functional Requirements

Functional requirements describe the system capabilities required for gastric disease detection and treatment recommendations.

Requirement ID	Functional Requirement Description
FR1	The system shall allow users to upload gastroscopy or endoscopy images for analysis.
FR2	The system shall receive symptom triggers from Component 1, including meal-related pain, spicy food triggers, and relief after antacid usage.
FR3	The system shall preprocess endoscopy images using image processing techniques such as resizing, normalization, and noise reduction.
FR4	The system shall analyze endoscopy images using a deep learning model (EfficientNet-B4) to detect gastric abnormalities.
FR5	The system shall classify detected gastric diseases such as gastritis, GERD, peptic ulcers, and hiatal hernia.
FR6	The system shall determine the severity level of the detected disease (Low, Medium, or Critical).
FR7	The system shall combine symptom data and image analysis results to confirm the gastric diagnosis.
FR8	The system shall generate an automated treatment recommendation based on the detected disease and severity level.
FR9	The system shall provide medication suggestions, including proton pump inhibitors, antacids, and lifestyle recommendations when applicable.
FR10	The system shall display the predicted disease type and severity level to the user.
FR11	The system shall highlight abnormal regions in endoscopy images using explainable AI visualization techniques.
FR12	The system shall store anonymized diagnostic results for research and system improvement purposes.

Table 4 : Functional Requirements

5.3.Non-Functional Requirements

Non-functional requirements describe the quality attributes that ensure the gastric diagnostic module operates effectively.

Security

The system must protect uploaded medical images and diagnostic results using secure authentication and access control mechanisms. Only authorized users should be able to access system functions.

Privacy

The system must ensure that patient data remains anonymous. Personally identifiable information, such as patient name, contact details, or identification numbers, should not be stored.

Performance

The system should process endoscopy images and generate diagnostic results within an acceptable response time to support clinical decision-making.

Accuracy

The deep learning model must achieve reliable classification performance when detecting gastric diseases from medical images to ensure trustworthy diagnostic support.

Usability

The system interface should be simple and easy to use so that medical students and healthcare trainees can upload images and understand diagnostic outputs without difficulty.

Explainability

The system should provide visual explanations for AI predictions by highlighting the image regions that contributed most to the disease classification.

Reliability

The system must consistently generate the same output for identical inputs and handle corrupted or incomplete image inputs gracefully.

Scalability

The system architecture should support future expansion, including integration with additional gastrointestinal diagnostic modules or advanced imaging techniques.

Maintainability

The system should allow developers to easily update deep learning models, treatment recommendation rules, and system components without requiring significant redesign.

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The proposed AI-assisted diagnostic system is primarily targeted at healthcare institutions, medical education institutes, and digital health technology providers. Hospitals and clinics that frequently manage patients with chest pain can benefit from the system by improving early diagnostic support and reducing unnecessary cardiac investigations. Medical universities and

training hospitals represent another key market segment because the system can function as an educational decision-support tool for medical students.

The primary customer persona includes medical students, clinical trainees, and healthcare professionals who require decision-support tools during diagnostic evaluation. Secondary users include hospital administrators and healthcare technology providers who seek AI-based solutions to improve diagnostic efficiency and reduce operational costs.

6.2. Value Proposition

The proposed system provides an intelligent clinical decision-support platform that assists in differentiating between cardiac-related and gastric-related chest pain. By integrating patient symptom analysis, cardiac diagnostic results, and gastric image analysis, the system helps improve diagnostic accuracy and supports faster clinical decision-making.

For Component 4 specifically, the Gastric Analysis and Decision Support Engine uses AI-based analysis of gastroscopy images combined with symptom data to identify gastric diseases such as gastritis, GERD, and peptic ulcers. The system also generates treatment recommendations and provides educational feedback for medical students. This combination of diagnostic support and learning assistance provides significant value for both clinical practice and medical education.

6.3. Revenue Model

The proposed system can be commercialized using several revenue models. One possible approach is a Software-as-a-Service (SaaS) model, where healthcare institutions subscribe to the platform on a monthly or yearly basis. Hospitals and clinics would pay subscription fees to access AI diagnostic tools, patient data management features, and medical imaging analysis services.

Another potential revenue stream is licensing the software to medical universities and training hospitals as an educational platform for clinical training. In addition, partnerships with digital health companies and medical technology providers may allow integration of the system into existing healthcare platforms.

6.4. Cost Estimation

The initial development costs of the system include expenses related to software development, AI model training, cloud infrastructure, and dataset acquisition. Hardware costs may include GPU-enabled computing resources for deep learning model training and storage servers for medical image datasets.

Operational costs include system maintenance, cloud hosting services, database management, and software updates. Additional costs may arise from data security implementation and regulatory compliance requirements related to healthcare technology systems.

6.5. Pricing Strategy

The pricing strategy for the proposed system can be based on a subscription-based model. Healthcare institutions may be offered tiered pricing plans depending on the number of users

and the level of system access required. For example, small clinics may subscribe to a basic diagnostic support package, while larger hospitals may use a full-featured enterprise version that includes advanced analytics and integration with hospital information systems.

For medical universities, an educational licensing model may be applied where institutions pay a yearly fee to provide system access for students and academic staff.

6.6.Competitive Advantage

The proposed system offers several competitive advantages compared to existing AI diagnostic tools. Many current systems focus on either cardiac disease detection or gastrointestinal disease analysis independently. In contrast, this system integrates multiple diagnostic components into a single decision-support framework capable of distinguishing between cardiac and gastric causes of chest pain.

In addition, the Gastric Analysis and Decision Support Engine introduces AI-based gastroscopy image classification combined with automated treatment recommendation. The system also provides explainable AI features such as visual explanations of image predictions, which improve transparency and support medical education.

6.7.Intellectual Property Considerations

The intellectual property of the proposed system includes the AI algorithms, software architecture, and integrated diagnostic workflow developed during the research. These components may be protected through software copyrights and potential patent applications related to the AI-based diagnostic decision framework.

Data privacy and ethical considerations must also be addressed when handling medical data. All patient information used for model training and system testing must be anonymized and handled according to healthcare data protection regulations. Licensing agreements may also be used to protect proprietary algorithms and prevent unauthorized system use.

7. BUDGET AND JUSTIFICATION

7.1.Estimated Budget Breakdown

Budget Category	Description	Estimated Cost (LKR)
Hardware	Personal laptop usage and minor hardware resources for developing and testing the gastric disease detection model	25,000
Software	Development tools, IDE plugins, and version control support tools used during system development	5,000
Cloud Services	Cloud server hosting for AI model deployment and backend API testing	30,000

Data Processing	Dataset preparation, annotation, preprocessing, and configuration of experiments for endoscopy images	15,000
Human Effort	Development, system testing, model training, and research implementation activities	20,000
Miscellaneous	Internet usage, documentation preparation, and research-related materials	5,000

Table 5 : Budget Breakdown for the Gastric Analysis and Decision Support Engine

Total Estimated Budget: 100,000 LKR

7.2. Hardware Cost Justification

The development of the Gastric Analysis & Decision Support Engine requires computing resources to train and test deep learning models for gastric disease detection using gastroscopy images.

The development tasks include:

- Training the **EfficientNet-B4** deep learning model
- Performing image preprocessing using OpenCV and Albumentations
- Running model evaluation experiments
- Testing backend APIs and system integration

Instead of purchasing new hardware, the research will utilize an existing personal laptop with adequate processing power and memory. Therefore, only a portion of the budget is allocated for hardware usage and potential upgrades such as additional storage devices or minor peripherals.

This approach significantly reduces development costs while still enabling efficient model training and system testing.

7.3. Software Requirements

Software Tool	Purpose	Cost
Python	Main programming language for AI model development	Free
PyTorch	Deep learning framework for EfficientNet-B4 implementation	Free
OpenCV / Albumentations	Image preprocessing and augmentation of endoscopy images	Free
FastAPI	Backend API development for connecting the AI model with the user interface	Free

PostgreSQL	Database management for storing system results and logs	Free
Git / GitHub	Version control and collaboration for project development	Free
Development Tools (IDE Support)	Optional development plugins and debugging tools	5,000

Table 6 : Software Requirements for System Development

Justification:

Most technologies used in this research are open-source tools widely adopted in artificial intelligence and medical imaging applications. These technologies significantly reduce software licensing costs while providing powerful capabilities for building AI-based diagnostic systems.

7.4.Cloud Services

Cloud Service	Purpose	Estimated Cost (LKR)
Cloud Server (AWS / GCP / Azure)	Hosting backend APIs and AI models	18,000
Cloud Storage	Storage for gastroscopy datasets and experiment results	7,000
Deployment Services	Container deployment and server configuration	5,000

Table 7 : Cloud Infrastructure Cost

Justification:

Cloud services enable remote deployment of the AI model and allow the system to be accessed through a web-based interface. Cloud infrastructure improves system scalability, reliability, and accessibility for users such as medical students and researchers.

7.5.Human Effort Cost

Human effort represents the time, expertise, and research activities required to design, implement, evaluate, and validate the Gastric Analysis & Decision Support Engine.

The development process includes several activities such as designing the symptom analysis workflow, implementing the EfficientNet-B4 model for gastric disease detection, developing backend APIs, integrating the system components, and performing testing and validation.

Activity	Description	Estimated Cost (LKR)
System Design	Designing gastric symptom analysis workflow and system architecture	4,000
AI Model Development	Training EfficientNet-B4 model for gastric disease detection	5,000
Backend Development	Implementing FastAPI backend services and database integration	3,000
Frontend Integration	Connecting system outputs with the user dashboard interface	2,000
System Testing & Validation	Evaluating model performance and system functionality	2,000
Research Travel	Transportation for consultations or clinical knowledge gathering	2,000
Documentation & Reporting	Preparation of research reports and system documentation	2,000

Table 8 : Human Effort Cost

7. WORK BREAKDOWN STRUCTURE (WBS)

8.1. Work Breakdown Structure Diagram

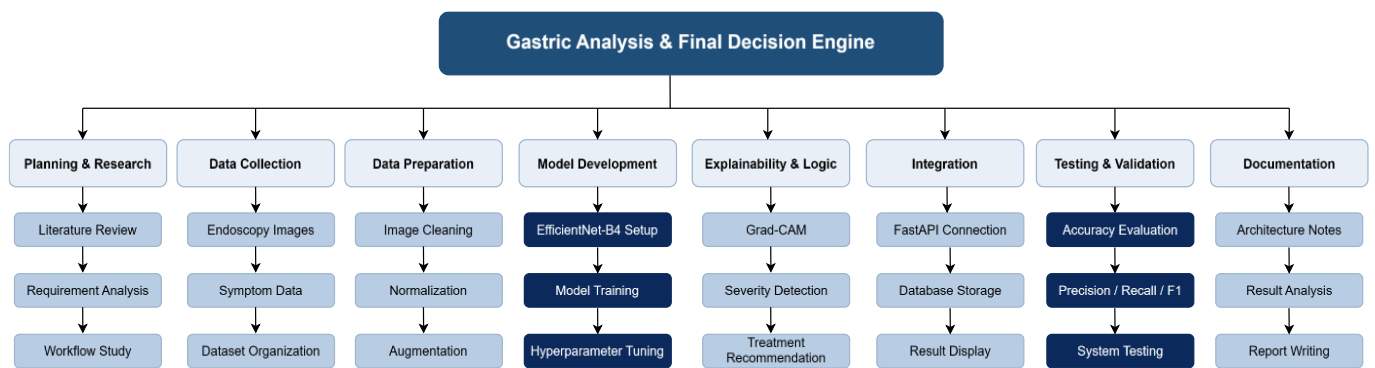


Figure 5 : WORK BREAKDOWN STRUCTURE

This diagram illustrates the hierarchical decomposition of the research project into smaller, manageable tasks. The project activities are divided among team members based on their respective system components and responsibilities.

8.2.Task Breakdown per Member

Task ID	Task Name	Description	Timeline
T1	Literature Review	Study research papers related to gastric disease diagnosis, endoscopy image analysis, and deep learning models such as EfficientNet used in medical imaging.	March 2026
T2	Dataset Collection	Collect gastroscopy / endoscopy image datasets related to gastric diseases such as gastritis, ulcers, GERD, and hiatal hernia from medical datasets (e.g., HyperKvasir).	March – April 2026
T3	Dataset Preparation & Annotation	Organize and label the collected endoscopy images according to disease categories and severity levels for supervised model training.	April 2026
T4	Image Preprocessing	Perform preprocessing tasks including image resizing, normalization, noise removal, and augmentation using OpenCV and Albumentations.	April – May 2026
T5	Gastric Disease Detection Model Development	Develop the deep learning model using EfficientNet-B4 implemented with PyTorch to classify gastric diseases from endoscopy images.	May – June 2026
T6	Disease Severity Classification	Implement classification logic to categorize detected gastric diseases into severity levels such as Low, Medium, and Critical.	June 2026
T7	Prescription Recommendation Engine	Develop a rule-based system that automatically generates treatment recommendations including medications, dosage suggestions, and dietary advice.	June – July 2026
T8	Backend API Development	Develop FastAPI backend services to process symptom data, handle endoscopy image inputs, and communicate with the trained AI model.	July – August 2026
T9	System Integration	Integrate the gastric analysis module with other system components and connect it with the central database and frontend interface.	August 2026
T10	System Testing and Validation	Evaluate model performance using metrics such as accuracy, precision, recall, and F1-score, and validate the overall system workflow.	September – October 2026

Table 9 : Task Breakdown per Member

8.3.Timeline Alignment

The development tasks for Gastric Analysis & Decision Support Engine are aligned with the official research project timeline to ensure that implementation, integration, and evaluation are completed before the final submission deadlines.

Phase	Period	Activities
Phase 1	March – April 2026	Literature review, dataset collection, and dataset annotation
Phase 2	April – May 2026	Image preprocessing and dataset preparation
Phase 3	May – June 2026	Gastric disease detection model development and severity classification
Phase 4	June – July 2026	Prescription recommendation engine development
Phase 5	July – August 2026	Backend API development and system integration
Phase 6	August – October 2026	System testing, evaluation, and optimization

Table 10 : Timeline Alignment

8.4.Balanced Workload Distribution

The workload for Component 4 is organized into several development stages to ensure balanced effort throughout the research project lifecycle.

The initial stage focuses on research and dataset preparation, including literature review, dataset collection, and annotation of endoscopy images. This stage ensures that sufficient high-quality medical image data is available for training the deep learning model.

The second stage focuses on image preprocessing and model development, where deep learning techniques such as EfficientNet-B4 are implemented to detect gastric diseases from endoscopy images. During this phase, disease classification and severity detection mechanisms are also developed.

The third stage involves system integration, including the development of backend APIs using Fast API and integration of the gastric analysis module with the overall diagnostic system.

The final stage focuses on system testing, evaluation, and optimization, ensuring that the model achieves acceptable performance levels and the entire workflow operates reliably before the final research submission.

This structured approach ensures that development progresses systematically from data preparation to AI model development, system integration, and evaluation.

9. GANTT CHART

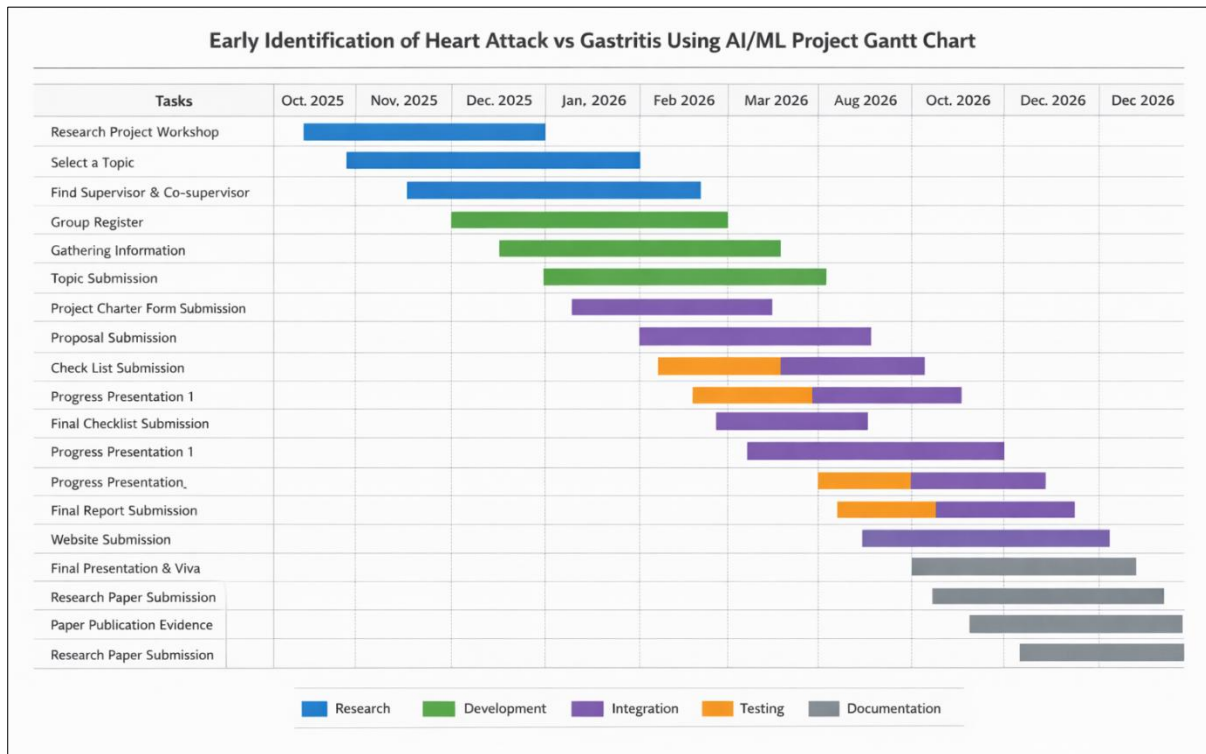


Figure 6 : Gantt Chart

References

- [1] A. J. Bösner, "Accuracy of symptoms and signs for coronary heart disease assessed in primary care," *British Journal of General Practice*, 2010.
- [2] J. E. Backus, "A prospective validation of the HEART score for chest pain patients at the emergency department," *International Journal of Cardiology*, vol. 168, no. 3, pp. 2153-2158, 2013.
- [3] R. Kumar, M. Singh and P. Pandey, "Hybrid machine learning approaches for cardiovascular disease prediction," *Computers in Biology and Medicine*, 2020.
- [4] H. Hannun and P. Rajpurkar, "Cardiologist-level arrhythmia detection with deep neural networks," *Nature Medicine*, vol. 25, 2019.
- [5] T. Aoki and T. Yamada, "Automatic detection of gastric cancer in endoscopic images using deep learning," *Gastric Cancer*, vol. 23, 2020.
- [6] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. W. M. van der Laak, B. van Ginneken and C. I. Sánchez, "A Survey on Deep Learning in Medical Image Analysis," *Medical Image Analysis*, vol. 42, no. 60-88, 2017.
- [7] H.-C. Shin, H. R. Roth, M. Gao, L. Lu, Z. Xu, I. Nogues, J. Yao, D. Mollura and R. M. Summers, "Deep Convolutional Neural Networks for Computer-Aided Detection in Medical Imaging," *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, p. 1285–1298, 2016.
- [8] A. Krizhevsky, I. Sutskever and G. Hinton, "ImageNet Classification With Deep Convolutional Neural Networks," in *Curran Associates, Inc.*, 2012.
- [9] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *International Conference on Machine Learning (ICML)*, Long Beach, USA, 2019.
- [10] A. Esteva, A. Robicquet, B. Ramsundar, V. Kuleshov, M. DePristo, K. Chou, C. Cui, G. Corrado, S. Thrun and J. Dean, "A Guide to Deep Learning in Healthcare," *Nature Medicine*, vol. 25, no. 1, pp. 24-29, 2019.
- [11] E. S. Berner, "Clinical decision support systems: State of the art," Rockville, MD, 2016.
- [12] Y. LeCun, Y. Bengio and G. Hinton, "Deep Learning," *Nature*, vol. 521, no. 7553, p. 436–444, 2015.

10.APPENDICES

Appendix A – Survey Instruments

This appendix contains the questionnaire used to collect patient symptom data related to chest pain and gastric conditions. The survey includes questions regarding patient demographics, lifestyle factors, meal-related symptoms, medication history, and other relevant clinical indicators that help differentiate cardiac and gastric chest pain.

Appendix B – Preliminary UI Mockups

This appendix includes preliminary user interface designs for the medical student dashboard. The mockups demonstrate how users can upload gastroscopy images, review AI predictions, view severity levels, and access treatment recommendations generated by the system.

Appendix C – Risk Analysis

Potential Risk	Description	Mitigation Strategy
Limited Dataset Availability	Insufficient endoscopy images for training the AI model may affect model performance.	Use publicly available datasets such as HyperKvasir and apply data augmentation techniques.
Model Accuracy Issues	The AI model may produce incorrect predictions due to limited or imbalanced data.	Use validation techniques, improve dataset quality, and tune model parameters.
System Integration Problems	Difficulty integrating the gastric analysis module with other system components.	Use modular architecture and well-defined APIs for communication between components.
Data Privacy Concerns	Handling medical images and patient data may raise privacy issues.	Ensure all datasets are anonymized and implement secure data storage methods.
Hardware or Resource Limitations	Limited computing power may slow down model training and testing.	Use cloud-based computing resources and optimize model training processes.

Table 11 : Risk Analysis

Appendix D – Ethical Documents

This appendix includes ethical considerations related to the use of medical datasets and patient information. All patient data used in the research will be anonymized to protect privacy, and datasets will be used strictly for research and educational purposes in accordance with medical data protection guidelines.